Econometrics Chapter 8: Instrumental Variables and GMM

(1) Instrumental Variable

Reason for IV [Endogeneity]

1. Omitted Variables
2. Measurement error
3. Simultaneity [Reverse Causality]

Instrumental Variable, Z

- $cov(X, Z) \neq 0$
- $cov(Z, \varepsilon) = 0$ [Z is not an omitted Variable

- used to provide consistent estimator for β_1

IV Estimation

$$\beta_1^{iv} = \frac{\sum (Y_i - \bar{Y}) Z_i}{\sum (X_i - \bar{X}) Z_i} = \frac{\sum (Z_i - \bar{Z}) Y_i}{\sum (Z_i - \bar{Z}) X_i}$$

- Consistency (converge in probability)

$$\beta_1^{iv} = \frac{\sum (Z_i - \bar{Z}) Y_i}{\sum (Z_i - \bar{Z}) X_i}$$

$$= \frac{\sum (Z_i - \bar{Z})(\beta_0 + \beta_1 X_i + \varepsilon_i)}{\sum (Z_i - \bar{Z}) X_i}$$

$$= \beta_1 + \frac{\frac{1}{n}\sum (Z_i - \bar{Z}) \varepsilon_i}{\frac{1}{n}\sum (Z_i - \bar{Z}) X_i} \xrightarrow{P} \beta_1 + \frac{\overbrace{Cov(Z, \varepsilon)}^{0}}{\underbrace{Cov(X, Z)}_{\neq 0}} = \beta_1$$

(2) Method of Movements

Method of Movements

Setup: $Y = \beta_0 + \beta_1 X + \varepsilon$

- $Cov(\varepsilon, Z) = 0$
- $E(\varepsilon | Z) = 0$ and $E(\varepsilon) = 0$
- $E(\varepsilon) = E(Y - \beta_0 - \beta_1 X) = 0$
- $E(\varepsilon Z) = E((Y - \beta_0 - \beta_1 X)Z) = 0$

[A] $\frac{1}{n}\sum(Y_i - \hat{\beta}_0^{mm} - \hat{\beta}_1^{mm} X_i) = 0$

[B] $\frac{1}{n}\sum(Y_i - \hat{\beta}_0^{mm} - \hat{\beta}_1^{mm} X_i)Z_i = 0$

Solving:

From [A]

$$\bar{Y} - \hat{\beta}_0^{mm} - \hat{\beta}_1^{mm} \bar{X} = 0$$

$$\therefore \hat{\beta}_0^{mm} = \bar{Y} - \hat{\beta}_1^{mm} \bar{X}$$

Substituting into [B]

$$\sum(Y_i - (\bar{Y} - \hat{\beta}_1^{mm} \bar{X}) - \hat{\beta}_1^{mm} X_i)Z_i = 0$$

$$\sum[(Y_i - \bar{Y}) - \hat{\beta}_1^{mm}(X_i - \bar{X})]Z_i = 0$$

$$\sum(Y_i - \bar{Y})Z_i - \hat{\beta}_1^{mm}\sum(X_i - \bar{X})Z_i = 0$$

$$\therefore \hat{\beta}_1^{mm} = \frac{\sum(Y_i - \bar{Y})Z_i}{\sum(X_i - \bar{X})Z_i}$$

(3) Two-Stage Least Squares (2SLS)

Two-Stage Least Squares

Setup: Given X_i and Z_i, decompose X_i into two uncorrelated parts by regressing X_i

$$X_i = \hat{\delta}_0 + \hat{\delta}_1 Z_i + r_{i\,xlz}$$

$$= \hat{X}_i + r_{i\,xlz}$$

- $\hat{X}_i$ is a linear function of Z_i

$E(\varepsilon Z_i) = 0$ then $E(\varepsilon \hat{X}_i) = 0$

- Movements in X_i that are correlated with ε_i are concentrated into $r_{i\,xlz}$

(Stage 1) Regress X_i on Z_i, compute $\hat{X}_i$

(Stage 2) Regress Y_i on $\hat{X}_i$ instead of X_i

Solving:

$$Y_i = \hat{\beta}_0^{2sls} + \hat{\beta}_1^{2sls} \hat{X}_i + \varepsilon_i$$

$$\underset{\hat{\beta}_0, \hat{\beta}_1}{\text{argmin}} : \Sigma(Y_i - \hat{\beta}_0^{2sls} - \hat{\beta}_1^{2sls} \hat{X}_i)^2$$

FOC: $-2\Sigma(Y_i - \hat{\beta}_0^{2sls} - \hat{\beta}_1^{2sls} \hat{X}_i) = 0$

$$-2\hat{X}_i \Sigma(Y_i - \hat{\beta}_0^{2sls} - \hat{\beta}_1^{2sls} \hat{X}_i) = 0$$

$$\hat{\beta}_0^{2sls} = \bar{Y} - \hat{\beta}_1^{2sls} \bar{X}$$

$$\hat{\beta}_1^{2sls} = \frac{\Sigma(Y_i - \bar{Y})(\hat{X}_i - \bar{\hat{X}}_i)}{\Sigma(\hat{X}_i - \bar{X})^2}$$

Expanding on $\hat{\beta}_1^{2sls}$

$$\hat{X}_i = \hat{\delta}_0 + \hat{\delta}_1 Z_1$$

$$\bar{\hat{X}}_i = \hat{\delta}_0 + \hat{\delta}_1 \bar{Z}$$

$$\hat{X}_i - \bar{\hat{X}}_i = \hat{\delta}_1 (Z_i - \bar{Z})$$

$$\Sigma(\hat{X}_i - \bar{\hat{X}}_i)^2 = \hat{\delta}_1^2 \Sigma(Z_i - \bar{Z})^2$$

Substitute back

$$\hat{\beta}_1^{2sls} = \frac{\hat{\delta}_1 \sum (z_i - \bar{z})(y_i - \bar{y})}{\hat{\delta}_1^2 \sum (z_i - \bar{z})^2} = \frac{\sum (z_i - \bar{z})(y_i - \bar{y})}{\hat{\delta}_1^2 \sum (z_i - \bar{z})^2}$$

$$\hat{\delta}_1 = \frac{\sum (z_i - \bar{z})(x_i - \bar{x})}{\sum (z_i - \bar{z})^2}$$

Therefore,

$$\hat{\beta}_1^{2sls} = \frac{\sum (z_i - \bar{z})(y_i - \bar{y})}{\sum (z_i - \bar{z})(x_i - \bar{x})}$$

(4) Overall IV

1. Demand & Supply

$Q^d = \delta_0 + \delta_1 P + \varepsilon^d$

$Q^s = \alpha_0 + \alpha_1 P + \alpha_2 R + \varepsilon^s$

$Q^s = Q^d$

- suppose R is observable and shifts supply but not demand function
  - $Cov(R, \varepsilon_t^d) = 0$
  - R shifts supply, supply shift changes prices
    $Cov(R, P) \neq 0$
  - R is a valid instrument for P in the demand eq^n $(Q = \alpha + \beta P + u)$

2. Consistency

First Argument: $\hat{\beta}_1^{IV} = \frac{\sum (z_i - \bar{z}) Y_i}{\sum (z_i - \bar{z}) X_i}$

$$= \frac{\sum (z_i - \bar{z})(\beta_0 + \beta_1 X_1 + \varepsilon_i)}{\sum (z_i - \bar{z}) X_i}$$

$$= \frac{\beta_0 \sum (z_i - \bar{z}) + \beta_1 \sum (z_i - \bar{z}) X_i + \sum (z_i - \bar{z}) \varepsilon_i}{\sum (z_i - \bar{z}) X_i}$$

$$= \beta_1 + \frac{\sum (z_i - \bar{z}) \varepsilon_i}{\sum (z_i - \bar{z}) x_i} \xrightarrow{p} \beta_1 + \frac{Cov(z, \varepsilon)}{Cov(z, x)} = \beta_1$$

Second Argument: $\beta_1 = \frac{Cov(z, \varepsilon)}{Cov(z, x)}$

$$\hat{\beta}_1 = \frac{\sum (z_i - \bar{z}) Y_i}{\sum (z_i - \bar{z}) X_i} \xrightarrow{p} \beta_1 + \frac{Cov(z, Y)}{Cov(z, x)} = \beta_1$$

3. Biasness

$$\hat{\beta}_1^{IV} = \beta_1 + \frac{\sum (z_i - \bar{z}) \varepsilon_i}{\sum (z_i - \bar{z}) X_i}$$

$$E(\hat{\beta}_1^{IV} | X \ldots, z \ldots) = \beta_1 + \frac{\sum (z_i - \bar{z}) E(\varepsilon_i | X \ldots, z \ldots)}{\sum (z_i - \bar{z}) X_i}$$

($E(\varepsilon_i | X \ldots, z \ldots) \neq 0$)

$\therefore \hat{\beta}_1^{IV}$ is biased

4. Standard Error

- assume homoskedastic error: $Var(\varepsilon | X..., Z...) = \sigma^2$

$$Var(\hat{\beta}_1^{IV} | X..., Z...) = Var\left(\beta + \frac{\sum (Z_i - \bar{Z})\varepsilon_i}{\sum (Z_i - \bar{Z}) X_i} \,\middle|\, X..., Z... \right)$$

$$= \frac{\sum (Z_i - \bar{Z})^2 \, Var(\varepsilon_i | X..., Z...)}{[\sum (Z_i - \bar{Z}) X_i]^2}$$

$$= \frac{\sum (Z_i - \bar{Z})^2 \sigma^2}{[\sum (Z_i - \bar{Z}) X_i]^2} \cdot \frac{\sum (X_i - \bar{X})^2}{\sum (X_i - \bar{X})^2}$$

$$= \frac{\sigma^2}{\sum (X_i - \bar{X})^2} \cdot \frac{\sum (z_i - \bar{z})^2 \sum (X_i - \bar{X})^2}{[\sum (Z_i - \bar{Z})(X_i - \bar{X})]^2}$$

$$= \frac{\sigma^2}{\sum (X_i - \bar{X})^2} \cdot \frac{1}{R_{XIZ}} = \frac{\sigma^2}{\sum (X_i - \bar{X})^2 R_{XIZ}} > Var(\hat{\beta}^{ols} | ...) = \frac{\sigma^2}{\sum (X_i - \bar{X})^2}$$

since $\frac{1}{R_{XIZ}}$ is a decimal

Note: Correlation btwn Z and X

$$R_{XIZ} = Z \text{ and } X = \frac{\sum (Z_i - \bar{Z})(X_i - \bar{X})}{\sqrt{\sum (z_i - \bar{z})^2 \sum (X_i - \bar{X})^2}}$$

$$R_{XIZ}^2 = \frac{[\sum (Z_i - \bar{Z})(X_i - \bar{X})]^2}{\sum (z_i - \bar{z})^2 \sum (X_i - \bar{X})^2}$$

5. Tradeoff for Consistent Estimator

- both OLS and IV are bias
- $Var(\hat{\beta}^{ols} | ...) < Var(\hat{\beta}^{iv} | ...)$
  - if $R_{XIZ} = 1$, back to SLR Z is a non valid instrument
  - if $R_{XIZ} \approx 0$, Z is a weak instrument, variance will be very large

6. Estimate of σ_{iv}^2

$$\hat{\sigma}_{iv}^2 = \frac{1}{n-2} \sum \hat{\varepsilon}_{i,iv}^2$$

$$R^2 = 1 - \frac{\sum (\hat{\varepsilon}_{i,iv}^2)}{\sum (Y_i - \bar{Y})^2}$$

(5) Matrix Algebra MM/2SLS/IV

a. MM

- sample moment conditions:

$\Sigma(Y_i - \hat{\beta}_0^{mm} - \hat{\beta}_1^{mm} X_i) = 0$ [A]

$\Sigma((Y_i - \hat{\beta}_0^{mm} - \hat{\beta}_1^{mm} X_i) Z_i) = 0$ [B]

- In Matrix:

[A] $y - X\hat{\beta}^{mm} = 0$

[B] $Z^T(y - X\hat{\beta}^{mm}) = 0$

$Z^T(y - X\hat{\beta}^{mm}) = Z^Ty - Z^TX\hat{\beta}^{mm} = 0$

$y = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}$ $X = \begin{bmatrix} 1 & X_1 \\ \vdots & \vdots \\ 1 & X_n \end{bmatrix}$ $+ \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{bmatrix}$ $Z = \begin{bmatrix} 1 & Z_1 \\ \vdots & \vdots \\ 1 & Z_n \end{bmatrix}$

$\therefore \hat{\beta}^{mm} = (Z^TX)^{-1}Z^Ty$

b. 2SLS

Step 1: X Regress on Z

$X = Z\beta + \varepsilon$

$\hat{\beta} = (Z^TZ)^{-1}Z^TX$

$\hat{X} = Z\hat{\beta} = Z(Z^TZ)^{-1}Z^TX$

Step 2: Regress Y on $\hat{X}$

$\hat{\beta}^{2sls} = (\hat{X}^T\hat{X})^{-1}\hat{X}^Ty$

$= [(Z(Z^TZ)^{-1}Z^TX)^T(Z(Z^TZ)^{-1}Z^TX)]^{-1}(Z(Z^TZ)^{-1}Z^TX)^Ty$

$= [X^TZ(Z^TZ)^{-1}Z^TZ(Z^TZ)^{-1}Z^TX]^{-1}(Z(Z^TZ)^{-1}Z^TX)^Ty$

$= [X^TZ(Z^TZ)^{-1}Z^TX]^{-1}(Z(Z^TZ)^{-1}Z^TX)^Ty$

$= (Z^TX)^{-1}(Z^TZ)(X^TZ)^{-1}(X^TZ(Z^TZ)^{-1}Z^Ty$

$= (Z^TX)^{-1}Z^Ty$

c. Testing for Consistency using Linear Algebra

$\hat{\beta}^{2sls} = \hat{\beta}^{mm} = \hat{\beta}^{iv}$

$= (Z^TX)^{-1}Z^Ty$

$= (Z^TX)^{-1}Z^T(X\beta + \varepsilon)$

$= (Z^TX)^{-1}Z^TX\beta + (Z^TX)^{-1}Z^T\varepsilon$

$= \beta + (Z^TX)^{-1}Z^T\varepsilon$

$= \beta + (\frac{1}{n}Z^TX)^{-1}(\underbrace{\frac{1}{n}Z^T\varepsilon}) \xrightarrow{p} \beta$

$\frac{1}{n}Z^T\varepsilon \xrightarrow{p} Cov(z, \varepsilon) = 0$

$\therefore$ Consistent

d. Standard Error & Variances – for asymptotically valid variance-covariance matrices

$\widehat{Var(\hat{\beta})} \xrightarrow{p} Var(\hat{\beta})$

Homoskedastic Error

$\hat{Var}(\hat{\beta}^{iv} | \dots) = Var(\beta + (Z^TX)^{-1}Z^T\varepsilon | \dots)$

$= (Z^TX)^{-1}Z^T\sigma^2 Z(X^TZ)^{-1}$

$= \sigma^2(Z^TX)^{-1}Z^TZ(X^TZ)^{-1}$

Heteroskedastic Error

$\hat{Var}(\hat{\beta}^{iv} | \dots) = (Z^TX)^{-1}(\sum \hat{\varepsilon}_i^2 z_{i*}^T z_{i*})(X^TZ)^{-1}$

(6) Scenarios Regressions with Exogenous & Endogenous Regressors, multiple IVs

Scenario 1: Over identified using MM

Suppose there is more IVs than endo regressors

Example: 1exo 1endo 2IV

$$Y = \beta_0 + \beta_1 X_1^k + \beta_2 X_2^g + \varepsilon$$

- X_i^k is exo
- X_2^g is endo

there exist z_2 and z_3 both correlated with X_2^g and uncorrelated with ε

Data Sample: $\{X_{i1}^k, X_{i2}^g, z_{i2}, z_{i3}, Y_i\}_{i=1}^n$

Sample Moments

$$\frac{1}{n}\sum \hat{\varepsilon}_i^{mm} = \frac{1}{n}\sum (Y_i - \hat{\beta}_0^{mm} - \hat{\beta}_1^{mm} X_{i1}^k - \hat{\beta}_2^{mm} X_{i2}^g) = 0$$

$$\frac{1}{n}\sum \hat{\varepsilon}_i^{mm} X_{i1}^k = \frac{1}{n}\sum (Y_i - \hat{\beta}_0^{mm} - \hat{\beta}_1^{mm} X_{i1}^k - \hat{\beta}_2^{mm} X_{i2}^g) X_{i1}^k) = 0$$

$$\frac{1}{n}\sum \hat{\varepsilon}_i^{mm} z_{i2} = \frac{1}{n}\sum (Y_i - \hat{\beta}_0^{mm} - \hat{\beta}_1^{mm} X_{i1}^k - \hat{\beta}_2^{mm} X_{i2}^g) z_{i2}) = 0$$

$$\frac{1}{n}\sum \hat{\varepsilon}_i^{mm} z_{i3} = \frac{1}{n}\sum (Y_i - \hat{\beta}_0^{mm} - \hat{\beta}_1^{mm} X_{i1}^k - \hat{\beta}_2^{mm} X_{i2}^g) z_{i3}) = 0$$

$\therefore$ 4 eqⁿ, 3 unknowns — cannot solve

Alternative solⁿ: Minimized sum of squared moments

$$(z^T y - z^T X \hat{\beta})^T (z^T y - z^T X \hat{\beta})$$

$$= (y^T z - \hat{\beta}^T X^T z)(z^T y - z^T X \hat{\beta})$$

$$= y^T z z^T y - y^T z z^T X \hat{\beta} - \hat{\beta}^T X^T z z^T y + \hat{\beta}^T X^T z z^T X \hat{\beta}$$

$$= y^T z z^T y - 2\hat{\beta}^T X^T z z^T y + \hat{\beta}^T X^T z z^T X \hat{\beta}$$

FOC wrt $\hat{\beta}^{mm}$: $-2X^T z z^T y + \hat{\beta}[X^T z z^T X + X^T z z^T X] = 0$

$$-X^T z z^T y + \hat{\beta} X^T z z^T X = 0$$

$$\hat{\beta}^{mm} = (X^T z z^T X)^{-1} (X^T z z^T y)$$

Testing for Consistency

$$\hat{\beta}^{mm} = (X^T Z Z^T X)^{-1}(X^T Z Z^T (X\beta + \varepsilon))$$

$$= (X^T Z Z^T X)^{-1}(X^T Z Z^T X\beta) + (X^T Z Z^T X)^{-1}(X^T Z Z^T \varepsilon)$$

$$= \beta + (X^T Z Z^T X)^{-1}(X^T Z Z^T \varepsilon)$$

$$= \beta + (X^T Z Z^T X)^{-1}\left(\frac{1}{n} X^T Z\right)\underbrace{\left(\frac{1}{n} Z^T \varepsilon\right)} \xrightarrow{p} \beta$$

$$\frac{1}{n} Z^T \varepsilon \xrightarrow{p} Cov(Z, \varepsilon) = 0$$

$\therefore \hat{\beta}^{mm}$ is consistent for overjustified

Standard Error

$$\hat{Var}(\hat{\beta}^{m} | \dots) = Var(\beta + (X^T Z Z^T X)^{-1}(X^T Z Z^T \varepsilon) | \dots)$$

$$= Var((X^T Z Z^T X)^{-1}(X^T Z Z^T \varepsilon) | \dots)$$

$$= (X^T Z Z^T X)^{-1}(X^T Z Z^T)\, Var(\varepsilon | \dots)(Z Z^T X)(X^T Z Z^T X)^{-1}$$

Homoskedasticity:

$$\hat{Var}(\hat{\beta}^{mm} | \dots) = \sigma^2 (X^T Z Z^T X)^{-1}(X^T Z Z^T Z Z^T X)(X^T Z Z^T X)^{-1}$$

Heteroskedasticity:

$$\hat{Var}(\hat{\beta}^{mm} | \dots) = (X^T Z Z^T X)^{-1}(X^T Z \Sigma \underset{mm}{\hat{\varepsilon}_i^2} Z_{i*}^T Z_{i*} Z^T X)(X^T Z Z^T X)^{-1}$$

Scenario 2: Over-identified 2SLS

Step 1: Regress X on Z

$$\hat{X} = Z\hat{B}$$

$$\hat{B} = (Z^TZ)^{-1}Z^TX$$

$$\hat{X} = Z(Z^TZ)^{-1}Z^TX$$

Step 2: Regress y on $\hat{X}$

$$\hat{\beta}^{iv} = (\hat{X}^T\hat{X})^{-1}\hat{X}^Ty$$

$$= [(Z(Z^TZ)^{-1}Z^TX)^T(Z(Z^TZ)^{-1}Z^TX)]^{-1}[(Z(Z^TZ)^{-1}Z^TX)^Ty]$$

$$= [(X^TZ(Z^TZ)^{-1}Z^T)(Z(Z^TZ)^{-1}Z^TX)]^{-1}[(Z(Z^TZ)^{-1}Z^TX)^Ty]$$

$$= [(X^TZ(Z^TZ)^{-1}Z^TX)]^{-1}[(Z(Z^TZ)^{-1}Z^TX)^Ty]$$

$$= (Z^TX)^{-1}(Z^TZ)\underbrace{(X^TZ)^{-1}}(X^TZ)(Z^TZ)^{-1}Z^Ty$$

X^TZ is not a square, cannot invert

not full rank - 4x3

- Follow up by matrix projection for solution

Consistency

$$\hat{\beta}^{2SLS} = (Z^TX)^{-1}(Z^TZ)(X^TZ)^{-1}(X^TZ)(Z^TZ)^{-1}Z^T(X\beta+\varepsilon)$$

$$= \beta + (Z^TX)^{-1}(Z^TZ)(X^TZ)^{-1}(X^TZ)(Z^TZ)^{-1}Z^T\varepsilon$$

$$= \beta + (\tfrac{1}{n}Z^TX)^{-1}(Z^TZ)(X^TZ)^{-1}(X^TZ)(Z^TZ)^{-1}\underbrace{(\tfrac{1}{n}Z^T\varepsilon)}$$

$\xrightarrow{p} \beta$

$\downarrow p$ $Cov(Z, \varepsilon) = 0$

Note: Overjustified case: $\hat{\beta}^{mm} \neq \hat{\beta}^{2SLS}$

(7) General Method of Moments (GMM)

GMM: Minimize weighted sum of squares

$$\hat{\beta}_W^{gmm} = \underset{\hat{\beta}}{\operatorname{argmin}} \underbrace{(z^Ty - z^TX\hat{\beta})^T W (z^Ty - z^TX\hat{\beta})}_{J(\omega)}$$

where W is some symmetric positive definite weight matrix

- X is $n \times (K+G+1)$
- Z is $n \times (K+M+1)$

Minimizing $J(\omega)$

$$\begin{aligned} J(\omega) &= (z^Ty - z^TX\hat{\beta})^T W (z^Ty - z^TX\hat{\beta}) \\ &= (z^Ty)^T - \hat{\beta}^T X^T z) W (z^Ty - z^TX\hat{\beta}) \\ &= (z^Ty)^T W (z^Ty - z^TX\hat{\beta}) - (\hat{\beta}^T X^T z) W (z^Ty - z^TX\hat{\beta}) \\ &= (z^Ty)^T W (z^Ty) - (z^Ty)^T W (z^TX\hat{\beta}) - (\hat{\beta}^T X^T z) W z^Ty + \hat{\beta}^T X^T z W z^T X \hat{\beta} \end{aligned}$$

FOC wrt $\hat{\beta}^{GMM}$: $-2X^TzWz^Ty + 2(X^TzWz^TX)\hat{\beta} = 0$

$$\hat{\beta}^{GMM} = (X^TzWz^TX)^{-1} X^TzWz^Ty$$

MM is GMM when $W = I_n$

2SLS is GMM when $W = (z^Tz)^{-1}$

Consistency

$$\begin{aligned} \hat{\beta}^{GMM} &= (X^TzWz^TX)^{-1} X^TzWz^Ty \\ &= (X^TzWz^TX)^{-1} X^TzWz^T(X\beta + \varepsilon) \\ &= \beta + (X^TzWz^TX)^{-1} X^TzWz^T\varepsilon \\ &= \beta + \left[\left(\tfrac{1}{n}X^Tz\right)W(z^TX)\right]^{-1} \left[\left(\tfrac{1}{n}X^Tz\right)W\underbrace{\left(\tfrac{1}{n}z^T\varepsilon\right)}_{\frac{1}{n}z^T\varepsilon \xrightarrow{p} cov(z,\varepsilon) = 0}\right] \xrightarrow{p} \beta \end{aligned}$$

Variance

- Homoskedasticity $= \sigma^2 (X^TzWz^TX)^{-1} X^TzWz^TzWz^TX (X^TzWz^TX)^{-1}$
- Heteroskedasticity $= (X^TzWz^TX)^{-1} X^TzW\Sigma\hat{\varepsilon}_i^2 z_{i*}^T z_{i*} Wz^TX (X^TzWz^TX)^{-1}$

Optimal Weight W

$$W=\left(\sum \hat{\varepsilon}_i^2{}_{gmm}\, z_{i*}^T z_{i*}\right)^{-1}$$

Variance under optimal weight:

- Homoskedasticity

$$\widehat{Var}(\hat{\beta}^{gmm}) = \hat{\sigma}^2 (X^T Z (Z^T Z)^{-1} Z^T X)^{-1}$$

where $\hat{\sigma}^2 = \frac{1}{n}\sum \hat{\varepsilon}_i^2{}_{gmm}$

- Heteroskedasticity

$$\widehat{Var}(\hat{\beta}^{gmm}) = \left(X^T Z \left(\sum \hat{\varepsilon}_{igmm}\, z_{i*}^T z_{i*}\right)^{-1} Z^T X\right)^{-1}$$

∴ Note: Form of variance of optimal GMM estimator under homoskedasticity is the same as 2SLS when $W=(Z^TZ)^{-1}$ in 2SLS

- 2SLS is as good as optimal GMM under homoskedasticity
- $\hat{\beta}$ not numerical identical but both asymptotically efficient for both 2SLS & Optimal GMM

(8) Inference after GMM Estimation

1. Testing Significance of β

a. Single Restriction

- do a t test

b. Multiple Restrictions (Asymptotic Chi-square test JF)

- Step 1: Formulate Wald Statistics

$$W = (R\hat{\beta}_{gmm} - r)^T (R\,Var(\hat{\beta}_{gmm})K^T)^{-1}(R\hat{g}_{mm} - r) \sim \chi^2_{(J)}$$

whereby $H_0: R\beta = r$ where R is $J \times K$ and r_0 is $K \times 1$

Step 2: Chi-square test JF / F-test

$$W \xrightarrow{d} \chi^2_{k-1}$$

$$p\text{-value} = P(\chi^2_{k-1} > W)$$

Reject H_0 if $p < 0.05$

2. Testing for Weak Instruments

Weak Instruments

- poorly correlated with the endogenous regressors
- result in poor finite sample properties
  - High Variances
  - large finite sample biases

Test:

Step 1: Regress each endogenous regressor on all exogenous regressors and instruments

Step 2: F-test on the significance of instruments in the first stage regression

Rule of thumb

$F > 20$ suggests instruments are strong

3. Tests of Overidentifying Restrictions

- checks overidentification and if instruments are valid

Overidentifying Case

- GMM Objective Function

$$J(w) = (z^T y - z^T x \hat{\beta})^T W (z^T y - z^T x \hat{\beta}) \neq 0$$

- GMM Estimator

$$\hat{\beta}^{GMM} = (z^T w)^{-1} \underbrace{(x^T z w)^{-1} (x^T z w)} z^T y$$

cannot cancel out since $x^T z$ is not full rank

Just identified

- Obj Function

$$J(w) = (z^T y - z^T x \hat{\beta}^{mm})^T W (z^T y - x \hat{\beta}^{mm}) = 0$$

given,

- Estimator

$$\hat{\beta}^{GMM} = (z^T x)^{-1} z^T y$$

and $z^T y - z^T x (z^T x^{-1} z^T y) = 0$

Hansen J-test

Step 1: Form $J(w)$ Null: All instruments are valid

- just $J(w) \approx 0$
- over $J(w) \neq 0$

Step 2: $J(w) \xrightarrow{d} \chi^2_{M-G}$ M-G is no. of excess instruments

Step 3: P-value

$p = P(\chi^2_{M-G} > J)$

Significant Result (Null Rejected)

(1) One or more of presume exo regressor is endo

(2) One or more instruments is not exo

(3) Some combo of the situations

4. Testing Endogeneity

- if there exist valid instruments, it allows us to test if one or more of the endogenous regressors can be treated as exogenous.
- Model: $Y = X\beta + \varepsilon$
  - $X = [1_n \; X^k_{*1} \dots X^k_{*k} \; X^g_{*k+1} \dots X^g_{*k+G}]$
  - $Z = [1_n \; X^k_{*1} \dots X^k_{*k} \; Z_{*k+1} \dots Z_{*,k+M}]$
- Population moment conditions:
  $E(Z^T\varepsilon) = 0$
- To test if X^g_{k+1} is not endo, add it to vector Z
  $\tilde{Z} = [1_n \; X^k_{*1} \dots X^k_{*k} \; X^g_{k+1} \; Z_{*k+1} \dots Z_{*,k+M}]$
  whereby $E(\tilde{Z}^T\varepsilon) = 0$

Test:

Step 1: Est the regression eq^n using instrument Z and get J_Z

Step 2: Est the regression eq^n using instrument $\tilde{Z}$ and get $J_{\tilde{Z}}$

Step 3: $C = J_{\tilde{Z}} - J_Z \xrightarrow{d} \chi^2(Q)$ Q - no. of endo variable tested for endogeneity

- X^g_{k+1} is exo $J_Z \approx J_{\tilde{Z}}$ $H_0: C = 0$
- X^g_{k+1} is endo $J_{\tilde{Z}} >> J(Z)$ $H_1: C \neq 0$

Step 4: P-value

$p = P(\chi^2 > C)$